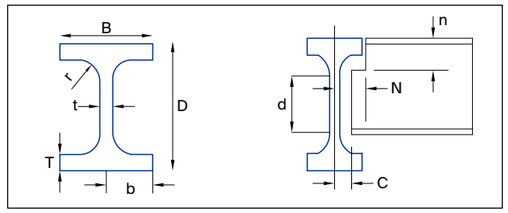
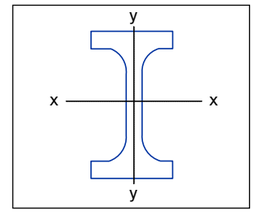
2.3 Universal Beams & Columns (UB & UC)



Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W4-4x4 (102x102)	13	19.4	105.7	103.1	8.8	7.1	6.4	75.5	24.7	5.88	10.6	6	58	15
	13.8	21.0	102.0	102.0	9.4	8.0	6.0	71.2	26.1	5.43	8.90	6	57	15
	16.3	24.0	107.0	100.0	12.0	7.9	6.0	71.0	30.9	4.17	8.99	6	56	18
W5-5x3 (127x76)	9	13.0	127.0	76.0	7.6	4.0	7.6	96.6	16.5	5.00	24.2	4	46	15
W5-5x5 (127x127)	16	23.8	127.3	127.0	9.1	6.1	7.6	93.8	30.4	6.95	15.4	5	70	17
	19	28.3	130.8	127.8	10.9	6.9	7.6	93.8	35.8	5.86	13.7	5	70	19
W6-6x3 (152x76)	9	14.0	150.0	75.0	7.0	5.0	8.0	120.0	17.8	5.36	24.0	5	45	15
	12	18.0	155.0	75.0	8.5	6.0	8.0	122.0	21.6	4.41	20.3	5	45	17
W6-6x3 1/2 (152x89)	11	16.0	152.4	88.7	7.7	4.5	7.6	121.8	20.3	5.76	27.1	4	52	15
W6-6x4 (152x102)	9	13.4	149.9	100.1	5.5	4.3	6.4	126.3	17.3	9.17	29.2	4	58	12
	12	17.9	153.2	101.6	7.1	5.8	6.4	126.3	22.9	7.14	21.6	5	58	13
	16	23.8	159.5	102.4	10.2	6.6	6.4	126.4	30.4	5.02	19.2	5	58	17
W6-6x6 (152X152)	15	22.3	152.1	152.1	6.6	5.8	6.4	126.2	28.5	11.5	21.6	5	83	13
	15.7	23.0	152.4	152.2	6.8	5.8	7.6	123.6	29.2	11.2	21.3	5	83	14
	20	30.0	157.6	152.9	9.4	6.5	7.6	123.6	38.3	8.13	19.0	5	83	17
	25	37.0	161.8	154.4	11.5	8.0	7.6	123.6	47.1	6.71	15.5	6	83	19
W7-7x 3 1/2 (178x89)	11	16.1	173.0	90.0	7.0	4.5	8.9	141.2	20.4	6.43	31.4	4	53	16
	12	18.1	175.0	90.0	8.0	5.0	8.9	141.2	23.0	5.63	28.2	5	53	17
	15	22.2	179.0	90.0	10.0	6.0	8.9	141.2	28.2	4.50	23.5	5	52	19
W7-7x4 (178x102)	13	19.0	177.8	101.2	7.9	4.8	7.6	146.8	24.3	6.41	30.6	4	58	16
W8-8x4 (203x102)	10	14.9	200.4	100.1	5.2	4.3	7.6	174.7	19.1	9.61	40.4	4	58	13
	13	19.4	202.9	101.6	6.5	5.8	7.6	174.7	24.8	7.84	29.9	5	58	14
	15	22.3	206.0	102.0	8.0	6.2	7.6	174.8	28.6	6.38	28.1	5	58	16
	16	23.1	203.2	101.8	9.3	5.4	7.6	169.4	29.4	5.47	31.4	5	58	17
W8-8x5 1/4 (203x133)	15	22.3	202.0	133.0	7.0	5.0	8.9	170.2	28.7	9.50	34.0	5	74	16
	17	25.1	203.2	133.2	7.8	5.7	7.6	172.4	32.0	8.54	30.2	5	74	15
	18	26.8	206.8	133.3	8.4	5.8	7.6	174.8	33.9	7.95	29.9	5	74	16
	20	30.0	206.8	133.9	9.6	6.4	7.6	172.4	38.2	6.97	26.9	5	74	17
	21	31.3	210.3	133.9	10.1	6.4	7.6	174.9	39.6	6.63	27.5	5	74	18

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
0.599	472	160	4.38	2.55	89.4	31.1	103	47.8	0.836	10.9	0.004	6.29
0.586	456	167	4.18	2.53	89.4	32.7	104	50.4	0.827	9.61	0.004	7.78
0.588	587	200	4.36	2.55	110	40.1	129	61.5	0.844	8.13	0.005	13.5
0.537	473	55.8	5.35	1.84	74.6	14.7	84.2	22.6	0.895	16.3	0.002	2.85
0.737	892	312	5.42	3.21	140	49.2	158	75.0	0.842	13.0	0.011	8.00
0.746	1092	380	5.52	3.25	167	59.4	190	90.6	0.845	11.2	0.014	13.1
0.576	666	49.5	6.11	1.67	88.8	13.2	102	20.8	0.876	20.4	0.003	2.81
0.584	841	60.2	6.24	1.67	109	16.0	126	25.4	0.875	17.6	0.003	4.81
0.638	834	89.8	6.41	2.10	109	20.2	123	31.2	0.890	19.6	0.005	3.56
0.681	683	91.4	6.29	2.30	91.1	18.3	102	28.1	0.873	26.2	0.005	1.69
0.690	918	125	6.33	2.33	120	24.5	136	38.0	0.869	20.4	0.007	3.76
0.704	1330	183	6.61	2.45	167	35.7	190	55.2	0.882	15.4	0.010	9.12
0.890	1210	387	6.51	3.68	159	50.9	177	77.7	0.837	21.4	0.021	4.21
0.889	1250	400	6.54	3.70	164	52.6	182	80.2	0.840	20.7	0.021	4.63
0.901	1748	560	6.76	3.83	222	73.3	248	112	0.849	16.0	0.031	10.5
0.912	2210	706	6.85	3.87	273	91.5	309	140	0.848	13.3	0.040	19.2
0.682	1060	85.4	7.20	2.04	123	19.0	138	29.4	0.884	23.9	0.006	3.15
0.685	1213	97.6	7.26	2.06	139	21.7	157	33.7	0.885	21.4	0.007	4.48
0.691	1529	122	7.36	2.08	171	27.1	195	42.3	0.886	17.8	0.009	8.16
0.738	1356	137	7.48	2.37	153	27.0	171	41.6	0.888	22.6	0.010	4.41
0.779	1284	87.3	8.19	2.14	128	17.4	145	27.2	0.867	36.3	0.008	1.78
0.787	1647	114	8.16	2.14	162	22.4	187	35.3	0.861	29.0	0.011	3.63
0.794	1999	142	8.36	2.23	194	27.8	222	43.7	0.869	25.2	0.014	5.68
0.790	2105	164	8.46	2.36	207	32.2	234	49.8	0.888	22.5	0.015	7.02
0.911	2105	275	8.56	3.09	208	41.3	232	63.4	0.879	27.9	0.026	4.50
0.915	2340	308	8.56	3.10	230	46.2	258	70.9	0.877	25.6	0.029	5.96
0.922	2578	331	8.72	3.12	249	49.7	279	76.3	0.880	24.5	0.033	7.13
0.923	2896	385	8.71	3.17	280	57.5	314	88.2	0.881	21.5	0.037	10.3
0.930	3119	405	8.87	3.20	297	60.4	333	92.7	0.885	21.0	0.041	11.6